BANDA SRI HARI

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GitHub | LinkedIn | Portfolio

EDUCATION

Sir M. Visvesvaraya Institute of Technology, Bengaluru — **Expected 2027**
B.E. in Artificial Intelligence & Machine Learning — **CGPA: 8.72/10**

EXPERIENCE

Chiacon Consulting Private Limited — **Mar 2026 – Jun 2026**
AI Engineer Intern — Remote, India

- Engineered a real-time voice pipeline using **Python, FastAPI WebSockets, and Azure Speech SDK**, streaming 16 kHz PCM audio with adaptive silence detection and software-level echo suppression.
- Integrated **Azure OpenAI and LangGraph** using Pydantic function-calling schemas, model routing, bounded conversational memory, and deterministic validation for reliable state-driven outputs.
- Integrated **Microsoft Graph, Entra ID, and Azure Communication Services** through REST APIs and OAuth 2.0/HMAC authentication, including a raw REST workaround for an Azure SDK limitation.
- Implemented **concurrent and multithreaded execution** using `asyncio.gather`, `threading.Thread`, `Timer`, `Event`, `Queue`, and thread-pool offloading for speech processing and parallel evaluation.
- Built a headless **7-persona regression harness** with 5 assertions; debugged race conditions, acoustic feedback, SDK failures, and file-write errors using defensive handling and graceful fallbacks.

SKILLS

Languages: Python, Java, SQL, JavaScript
Software Engineering: OOP, DSA, SDLC, REST APIs, FastAPI, Async Programming, Multithreading, Design Patterns, Unit Testing, Debugging, Git, GitHub Actions
Databases: PostgreSQL, Supabase, pgvector, SQLite, DynamoDB
Cloud & Web: AWS Lambda, API Gateway, S3, Amazon Bedrock, Azure OpenAI, Azure Speech, Microsoft Graph API, Azure Communication Services, WebSockets, SSE
AI/ML: Machine Learning, LangGraph, LangChain, RAG, Embeddings, Scikit-learn, XGBoost, LLM Evaluation
Frontend/Integration: React, Pydantic, FAISS, Tree-sitter/AST, Jira API, Slack API, GitHub API

PROJECTS

GitNova – Open-Source Contribution Intelligence Platform — Live | GitHub

- Engineered and deployed a full-stack code-intelligence platform using **React, FastAPI, Supabase PostgreSQL, and pgvector** to identify actionable GitHub issues and repository context.
- Designed PostgreSQL backend services with **16 migration scripts, relational constraints, GIN full-text search, pgvector similarity queries, PL/pgSQL RPCs, and asyncpg pooling** for REST APIs.
- Applied **OOP/polymorphism, Strategy/Adapter/Pipes-and-Filters patterns, AST parsing, Reciprocal Rank Fusion, and thread-safe synchronization** across deterministic retrieval and validation pipelines.
- Built **22 pytest suites and GitHub Actions automation** with fail-closed publication gates; debugged PostgreSQL/PostgREST, search, caching, and database-integrity failures.

Stack: Python, FastAPI, React, PostgreSQL, pgvector, Tree-sitter, GitHub Actions, RRF

Unshell – AML & KYB Forensic Intelligence Graph — *Hackfest 2026 – 2nd Runner-Up* | GitHub

- Architected an **8-node Python/FastAPI backend** integrating Companies House APIs, document ingestion, RAG, graph analysis, sanctions screening, and structured JSON outputs.
- Applied **data structures and algorithms** using directed graphs, Johnson's cycle detection, in/out-degree analysis, recursive Level-2 traversal, visited-set guards, and RapidFuzz entity verification.
- Engineered a **SQLite sanctions layer** with indexed schemas, batch loading, and parameterized asynchronous SQL queries via aiosqlite for OFAC screening.
- Built Pytest coverage for graph construction, cycle detection, score validation, and RAG verification; debugged API inconsistencies, graph failures, and LLM hallucinations with fail-closed validation.

Stack: Python, LangGraph, FastAPI, PyMuPDF, FAISS, Sentence-Transformers, RapidFuzz, NetworkX, SQLite

Orbit – Serverless AI Learning Platform — *AWS Hackathon*

- Engineered and deployed a serverless **FastAPI backend on AWS Lambda/API Gateway with Mangum**, implementing 8 REST endpoints and Pydantic schemas for curriculum, code evaluation, viva, session, and mentor workflows.
- Built **DynamoDB-backed session and assessment workflows** using composite keys and atomic updates, with Strategy/FSM patterns for evaluation routing and state transitions.
- Integrated **Amazon Bedrock, Knowledge Bases, Guardrails, and S3** into a RAG learning pipeline for transcript ingestion, checkpoint generation, and grounded mentor responses.
- Diagnosed live cloud failures involving **Lambda/API Gateway parsing, runtime exceptions, and Bedrock Guardrail false positives**; added safe parsing, fallbacks, and defensive recovery.

Stack: Python, FastAPI, AWS Lambda, API Gateway, DynamoDB, Amazon Bedrock, S3, React

ACHIEVEMENTS & CERTIFICATIONS

- **Hackfest 2026:** 2nd Runner-Up – Top 50 / 750+ teams
- **Amazon ML Challenge 2025:** Top 15% among 10,000+ teams
- **NPTEL:** Mathematical Foundations for Machine Learning – Elite Silver (IIT Guwahati)
- **Oracle Certified:** Generative AI Professional, Data Science Professional, Database@AWS Foundations Associate